Fiscal Year:	FY 2005	**Task Last Updated:**	FY 02/04/2005
PI Name:	Voorhees, Peter W. Ph.D.		
Project Title:	The Evolution of Dendrite Morphology during Isothermal coarsening		
Division Name:	Physical Sciences		
Program/Discipline:	RADIATION SHIELDING		
Program/Discipline--Element/Subdiscipline:	RADIATION SHIELDING--Radiation shielding		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	60208-0893	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-UNSOL
Start Date:	04/24/2000	**End Date:**	11/30/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG8-1660		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Lead-tin samples containing different volume fractions of dendritic tin-phase are prepared and coarsened for various times at 185°C. After the coarsening process the samples are serial-sectioned, which allows the three-dimensional reconstruction of the microstructure. Using the obtained three-dimensional images the microstructure is characterized with regard to the interface curvatures and the genus of the dendritic structure. The variations of the interface curvatures are understood as the source of the coarsening process. Overall, this process allows the study of the evolution of dendritic microstructures due to coarsening using an ideal alloy system. The objective of this research project is the first complete characterization of dendrite coarsening for varying volume fractions of dendritic phase. In the production process of most metal objects casting is one step. During the casting process, a liquid metal is solidified and the exact conditions of the casting process will determine the microstructure of the metal, which determines the final mechanical		

	properties of the ingot. When the microstructure is first formed at the beginning of the solidification dendrites are formed, which during the continuing process change shape due to coarsening. The coarsening process has a strong influence on the final microstructure and thus on the final mechanical properties. A fundamental understanding and characterization of this process will lead to improved simulations of casting processes. Improved simulations of casting processes will lead to improved casting processes and to better materials. Thus, our research is fundamentally important for progress in casting technology.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research will create a benchmark data set for the simulation of dendrite coarsening, which is a necessary part of simulations of casting processes
Task Progress:	We made significant progress this year. Specifically: We have shown that the morphological evolution of topologically complex systems, such as those found during dendritic solidification, during coarsening can be quantified by measuring the probability of finding a patch of interface with a given curvature tensor. We find that the morphological evolution is described by the flow of probability density in this curvature space that is induced by the coarsening process. The hallmark of our approach is a close coupling between experiment and theory; we use the experimentally measured three-dimensional microstructure as an input to a phase-field calculation that then determines the flow in curvature space. The methodology is general, and applicable to many systems undergoing coarsening,regardless of their topology. The coarsening process of dendritic microstructures was also studied in the Pb-Sn system using threedimensional reconstructions. The evolution of the microstructure during isothermal coarsening was analyzed in a system with a solid volume fraction of 58%. We determined the morphology of the microstructure by measuring the interfacial shape distribution, the probability of finding a patch of surface with a given pair of principle curvatures. In addition, we characterized the anisotropy of the microstructure through measurements of the probability of finding an interfacial normal in a certain direction. A linear increase of the inverse surface area per unit volume over the cube root of time is observed, despite the apparent lack of microstructural self-similarity. Interfacial normal distributions of the microstructures demonstrate a strong preferential directionality, specifically evolution to twofold symmetry, as coarsening time proceeds. During coarsening the fraction of interface that is nearly parallel to the directional solidification direction increases dramatically. This preferred direction exists because interfaces that are parallel to the directional solidification direction have lower curvature than the average at the earliest coarsening time. The three-dimensional morphological evolution of a directionally solidified Al–15 wt.% Cu alloy during isothermal coarsening was examined. The microstructure was analyzed using the Interfacial Shape Distribution (ISD), which gives the probability of locating an interfacial patch with a given pair of principle curvatures. By scaling the ISD's by a characteristic length, the inverse of the surface area per unit volume, it was possible to determine if the system reached a scaled time-invariant state. During coarsening, cylindrical interfaces appear in the second coarsened sample and remained for later coarsening times. The appearance of the corrresponding peak in the ISD indicated that the microstructure was not evolving in a self-similar manner, despite the linear relationship between the inverse of the surface area per unit volume and the cube root of coarsening time. Stereographic projections of the interfacial normals indicated that the microstructure of the earliest coarsened sample had a four-fold symmetry that disappeared during coarsening and that the majority of interfacial area for all coarsened samples is parallel to the direction in which the pre-coarsened samples were solidified.
Bibliography Type:	Description: (Last Updated: 01/17/2024)
Articles in Peer-reviewed Journals	Mendoza R, Alkemper J, Voorhees PW. "Three-dimensional morphological characterization of coarsened microstructures." Zeitschrift für Metallkunde. International Journal of Materials Research 2005 Feb;p. 155-160. http://www.ijmr.de/MK101013 , Feb-2005
Articles in Peer-reviewed Journals	Kammer D, Voorhees PW. "The morphological evolution of dendritic microstructures during coarsening." Acta Materialia.2006 Apr;54(6):1549-58. http://dx.doi.org/10.1016/j.actamat.2005.11.031 , Apr-2006
Articles in Peer-reviewed Journals	Mendoza R, Savin I, Thornton K, Voorhees PW. "Topological complexity and the dynamics of coarsening." Nature Materials. 2004 Jun;3(6):385-8. Epub 2004 May 23. PMID: 15156200 , Jun-2004
Books/Book Chapters	Mendoza R, Alkemper J, Voorhees PW. "The Topology of Coarsened Microstructures." in "Solidification Processes and Microstructures: A Symposium in Honor of Wilfried Kurz." Ed. M. Rappaz, C. Beckerman, R. Trivedi. TMS/Wiley, 2004. From 2004 TMS Annual Meeting held in Charlotte, North Carolina, March 14-18, 2004., Mar-2004
Presentation	Kammer, D.; Mendoza, R.; and Voorhees, P.W. "92. Topological Complexity and the Dynamic of Coarsening: New Insights via Three-Dimensional Reconstructions" Materials Laboratory, Wright Patterson Air Force Base Feb-2004
Presentation	Mendoza, R.; Kammer, D.; Thornton, K.; and Voorhees, P.W. "Coarsening of Dendritic Microstructures" Wilfried Kurz Symposium, TMS Annual meeting Mar-2004
Presentation	Savin, I.; Thornton, K.; and Voorhees, P.W. "95. Phase Field Simulations of Coarsening in Three Dimensional Topologically Complex Structures" TMS Annual meeting Charlotte NC Mar-2004
Presentation	Savin, I.; Thornton, K.; and Voorhees, P.W. "Coarsening in Topologically Complex Systems: Experiments and Simulations" SIAM Conference on the Mathematical Aspects of Materials Science, Los Angeles CA Apr-2004

Presentation	Kammer, D.; Mendoza, R.; and Voorhees, P.W. "Using Three-Dimensional Reconstructions to Characterize Coarsening in Topologically Complex Systems" TMS Fall meeting, New Orleans LA Sep-2004
Presentation	Kammer, D.; Mendoza, R.; Thornton, K.; and Voorhees, P.W. "10Coarsening in Topologically Complex Structures: Experiment and Theory" Materials Science and Engineering Department Colloquium, Iowa State University, Ames IA Oct-2004
Significant Media Coverage	Wu DT. "Coarsening dynamics: flowing complex shapes. Commentary on article by Mendoza R, Savin I, Thornton K, Voorhees PW. 'Topological complexity and the dynamics of coarsening.' Nature Materials. 2004 Jun;3(6):385-8. PMID: 15156200 " Nature Materials. 2004 Jun;3(6):353-4. http://dx.doi.org/10.1038/nmat1144 , Jun-2004